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EAS 4300 HW #7 Question 1

Stephen Miller

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A non-afterburning turbojet is being designed for operation at an altitude of 15 km and a Mach number of 1.8. The maximum stagnation temperature at the inlet of the turbine is 1500 K. The fuel is a jet fuel having a LHV of 43124 kJ/kg and f_{st} is 0.06. The following efficiencies apply at this Mach number: $\eta_d = 0.9$, $\eta_c = 0.9$, $\eta_b = 0.98$, $r_b = 0.97$, $\eta_t = 0.92$, $\eta_n = 0.98$. Use a γ value of 1.4 up to the burner, and a value of 1.3 for the rest of the engine. Assume R is 0.287 kJ/kgK throughout the engine. Plot the specific thrust, TSFC, η_{th} , η_p , and η_o as a function of rc , the total pressure ratio across the compressor. Is there an optimum rc that minimizes TSFC? Is there an optimum rc that maximizes specific thrust? Consider a range of rc from 2 to 60. Assume the exhaust is ideally expanded. Also plot the nozzle area ratio as a function of rc .

```
clc; clear; close all;
```

Given Parameters

M	= 1.8;	% Mach number
T04_max	= 1500;	% Maximum turbine inlet temperature
[K]		
delta_h_c	= 43124;	% Fuel lower heating value [kJ/kg]
f_st	= 0.06;	% Stoichiometric fuel-to-air ratio
eta_d	= 0.9;	% Diffuser efficiency
eta_c	= 0.9;	% Compressor efficiency
eta_b	= 0.98;	% Burner efficiency
eta_t	= 0.92;	% Turbine efficiency
eta_n	= 0.98;	% Nozzle efficiency
rb	= 0.97;	% Burner pressure ratio (P04/P03)
k	= [1.4 1.3];	% Specific heat ratios (before/after burner)
R	= 0.287;	% Gas constant [kJ/kg·K]
Ta	= 216.65;	% Ambient temperature at 15 km [K]
Pa	= 1.2112e4;	% Ambient pressure at 15 km [Pa]

```

delta    = 58;                                % Number of data points for rc
rc       = linspace(2, 60, delta)';          % Compressor pressure ratio range

```

Preallocate Variables

```

n        = length(rc);
T03s     = zeros(n, 1);
T03      = zeros(n, 1);
W        = zeros(n, 1);
f_b      = zeros(n, 1);
T04      = zeros(n, 1);
P04      = zeros(n, 1);
T05      = zeros(n, 1);
T05s     = zeros(n, 1);
P05      = zeros(n, 1);
T7s      = zeros(n, 1);
T7       = zeros(n, 1);
Me       = zeros(n, 1);
Ue       = zeros(n, 1);
I        = zeros(n, 1);
TSFC     = zeros(n, 1);
eta_th   = zeros(n, 1);
eta_p    = zeros(n, 1);
eta_0    = zeros(n, 1);
Ratio    = zeros(n, 1);

```

Inlet and Diffuser

Stagnation temperature and flight velocity:

$$T_{0a} = T_a \left(1 + \frac{\gamma_1 - 1}{2} M^2 \right), \quad u = M \sqrt{\gamma_1 R T_a}$$

The diffuser is adiabatic, so $T_{02} = T_{0a}$. Its efficiency sets the pressure recovery:

$$T_{02s} = T_a + \eta_d (T_{0a} - T_a), \quad P_{02} = P_a \left(\frac{T_{02s}}{T_a} \right)^{\gamma_1/(\gamma_1 - 1)}$$

Specific heats: $c_{p1} = \frac{\gamma_1}{\gamma_1 - 1} R$ for air ($\gamma_1 = 1.4$) and $c_{p2} = \frac{\gamma_2}{\gamma_2 - 1} R$ for burned gas ($\gamma_2 = 1.3$).

```

T0a      = Ta * (1 + ((k(1) - 1)/2) * M^2);          % T0a = T02
u        = M * sqrt(R * 1000 * k(1) * Ta);          % Flight
velocity [m/s]
T02s     = eta_d * (T0a - Ta) + Ta;                  % Isentropic-
equivalent T02
P02      = Pa * (T02s / Ta)^(k(1) / (k(1) - 1));      % Diffuser
exit pressure

```

```

% Specific heats [kJ/kg·K]: air (k = 1.4) and burned gas (k = 1.3)
Cp = (k ./ (k - 1)) * R;

```

Turbojet Cycle for Each rc

Compressor (efficiency η_c):

$$T_{03s} = T_{02} r_c^{(\gamma_1-1)/\gamma_1}, \quad T_{03} = T_{02} + \frac{T_{03s} - T_{02}}{\eta_c}, \quad w_c = c_{p1}(T_{03} - T_{02})$$

Main burner energy balance, with the stoichiometric limit $f_b \leq f_{st}$:

$$c_{p1}T_{03} + f_b \eta_b \Delta h_c = (1 + f_b) c_{p2}T_{04} \Rightarrow f_b = \frac{c_{p2}T_{04} - c_{p1}T_{03}}{\eta_b \Delta h_c - c_{p2}T_{04}}, \quad P_{04} = r_b r_c P_{02}$$

If f_b would exceed f_{st} , the burner runs at f_{st} and T_{04} is solved from the same balance.

Turbine work drives the compressor (efficiency η_t):

$$(1 + f_b) c_{p2}(T_{04} - T_{05}) = w_c, \quad T_{05s} = T_{04} - \frac{T_{04} - T_{05}}{\eta_t}$$

$$P_{05} = P_{04} \left(\frac{T_{05s}}{T_{04}} \right)^{\gamma_2/(\gamma_2-1)}$$

Nozzle expands ideally to P_a (efficiency η_n), with $(T_0, P_0) = (T_{05}, P_{05})$ and $T_e = T_7$:

$$T_{es} = T_0 \left(\frac{P_a}{P_0} \right)^{(\gamma_2-1)/\gamma_2}, \quad T_e = T_0 - \eta_n(T_0 - T_{es})$$

$$M_e = \sqrt{\frac{2}{\gamma_2-1} \left(\frac{T_0}{T_e} - 1 \right)}, \quad u_e = M_e \sqrt{\gamma_2 R T_e}$$

$$\frac{A_e}{A^*} = \frac{1}{M_e} \left[\frac{2}{\gamma_2+1} \left(1 + \frac{\gamma_2-1}{2} M_e^2 \right) \right]^{\frac{\gamma_2+1}{2(\gamma_2-1)}}$$

Performance, with $f = f_b$:

$$I = (1 + f) u_e - u, \quad \text{TSFC} = \frac{f}{I}$$

$$\eta_{th} = \frac{(1 + f) u_e^2 - u^2}{2f \Delta h_c}, \quad \eta_p = \frac{I u}{\frac{1}{2} [(1 + f) u_e^2 - u^2]}, \quad \eta_0 = \eta_{th} \eta_p$$

```
for j = 1:n
    % Compressor
    T03s(j) = T0a * rc(j)^((k(1) - 1) / k(1)); % Ideal compressor
    exit temperature
    T03(j) = ((T03s(j) - T0a) / eta_c) + T0a; % Actual compressor
    exit temperature
    W(j) = Cp(1) * (T03(j) - T0a); % Compressor work
```

[kJ/kg]

```
% Burner energy balance:  $C_p(1)T_{03} + f\eta_b dh_c = (1 + f)C_p(2)T_{04}$ 
f_b(j) = (Cp(2) * T04_max - Cp(1) * T03(j)) / (eta_b * delta_h_c - Cp(2)
* T04_max);
T04(j) = T04_max;
if f_b(j) > f_st
    % Stoichiometric limit: burn at f_st and find the lower T04
    f_b(j) = f_st;
    T04(j) = (Cp(1) * T03(j) + f_st * eta_b * delta_h_c) / ((1 + f_st) *
Cp(2));
end
P04(j) = rb * rc(j) * P02;

% Turbine drives the compressor (T05 = T06 = T07)
T05(j) = T04(j) - (W(j) / (Cp(2) * (1 + f_b(j))));
T05s(j) = ((T05(j) - T04(j)) / eta_t) + T04(j);
P05(j) = P04(j) * (T05s(j) / T04(j))^(k(2) / (k(2) - 1)); % P05 = P06

% Nozzle: ideal expansion to Pa
T7s(j) = T05(j) / (P05(j) / Pa)^((k(2) - 1) / k(2));
T7(j) = eta_n * (T7s(j) - T05(j)) + T05(j);
Me(j) = sqrt(((T05(j) / T7(j)) - 1) / ((k(2) - 1) / 2));
Ue(j) = Me(j) * sqrt(k(2) * R * 1000 * T7(j)); % Exhaust velocity
[m/s]

% Performance
I(j) = (1 + f_b(j)) * Ue(j) - u; % Specific thrust
[m/s]
TSFC(j) = f_b(j) / I(j); % Thrust-specific
fuel consumption [s/m]
eta_p(j) = (I(j) * u) / (((1 + f_b(j)) * (Ue(j)^2) / 2) - (u^2) / 2);
eta_th(j) = (((1 + f_b(j)) * Ue(j)^2 - u^2) / (2 * f_b(j) * delta_h_c *
1000));
eta_0(j) = eta_th(j) * eta_p(j); % Overall efficiency
Ratio(j) = (1 / Me(j)) * ((2 / (k(2) + 1)) * (1 + ((k(2) - 1) / 2) *
Me(j)^2))^(k(2) + 1) / (2 * (k(2) - 1)));
end
```

Optimum rc

```
[I_max, iI] = max(I);
[TSFC_min, iT] = min(TSFC);
fprintf('Maximum specific thrust: I = %.1f m/s at rc = %.2f\n', I_max,
rc(iI));
fprintf('Minimum TSFC: %.4e s/m at rc = %.2f\n', TSFC_min, rc(iT));
```

Maximum specific thrust: I = 708.2 m/s at rc = 6.07

Minimum TSFC: 3.5901e-05 s/m at rc = 42.70

Figure 1: Specific Thrust vs. r_c

```
figure('Position', [100, 100, 600, 400]);
plot(rc, I, 'LineWidth', 1.5);
hold on;
plot(rc(iI), I_max, 'o', 'MarkerSize', 6);
text(rc(iI) + 1, I_max, sprintf('max: (%.2f, %.1f)', rc(iI), I_max),
'FontSize', 8, 'VerticalAlignment', 'bottom');
hold off;
xlabel('r_c [dim]', 'FontWeight', 'bold', 'FontSize', 10);
ylabel('I [m/s]', 'FontWeight', 'bold', 'FontSize', 10);
title('Specific Thrust vs. r_c', 'FontSize', 12);
grid on;
```

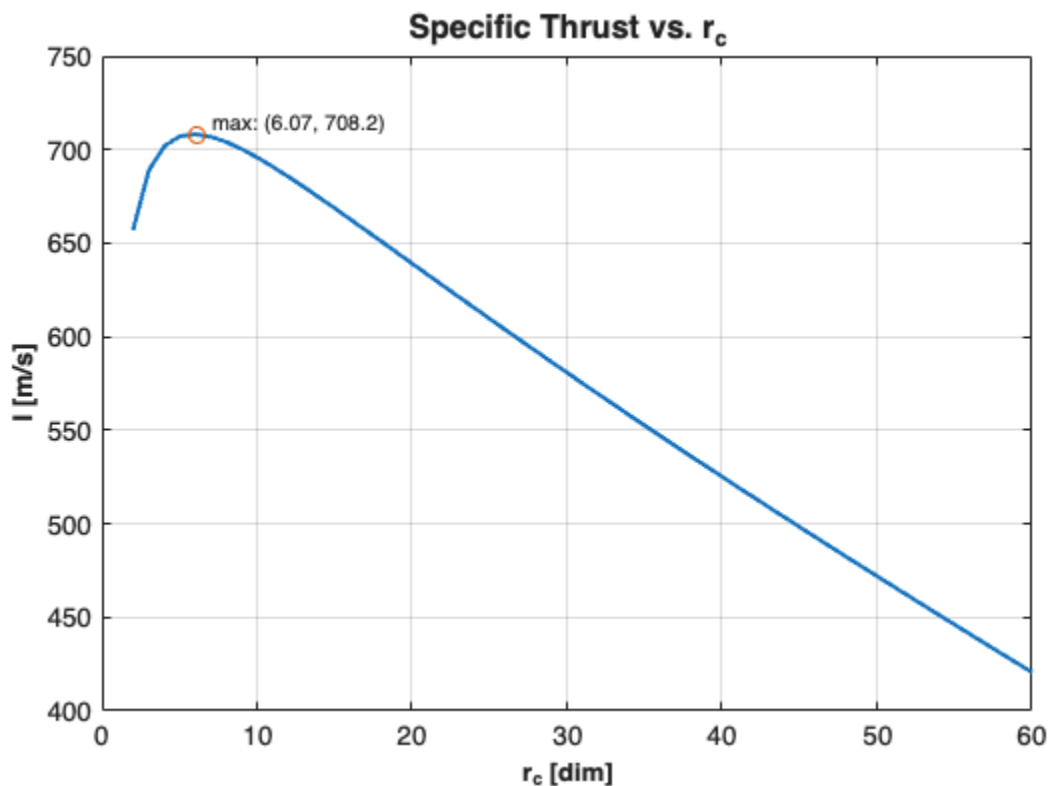


Figure 2: TSFC vs. r_c

```
figure('Position', [150, 150, 600, 400]);
plot(rc, TSFC, 'LineWidth', 1.5);
hold on;
plot(rc(iT), TSFC_min, 'o', 'MarkerSize', 6);
text(rc(iT), TSFC_min, sprintf('min: (%.2f, %.3e)', rc(iT), TSFC_min),
'FontSize', 8, ...
'VerticalAlignment', 'top', 'HorizontalAlignment', 'center');
hold off;
ax = gca;
ax.YAxis.Exponent = 0;
```

```
xlabel('r_c [dim]', 'FontWeight', 'bold', 'FontSize', 10);
ylabel('TSFC [s/m]', 'FontWeight', 'bold', 'FontSize', 10);
title('TSFC vs. r_c', 'FontSize', 12);
grid on;
```

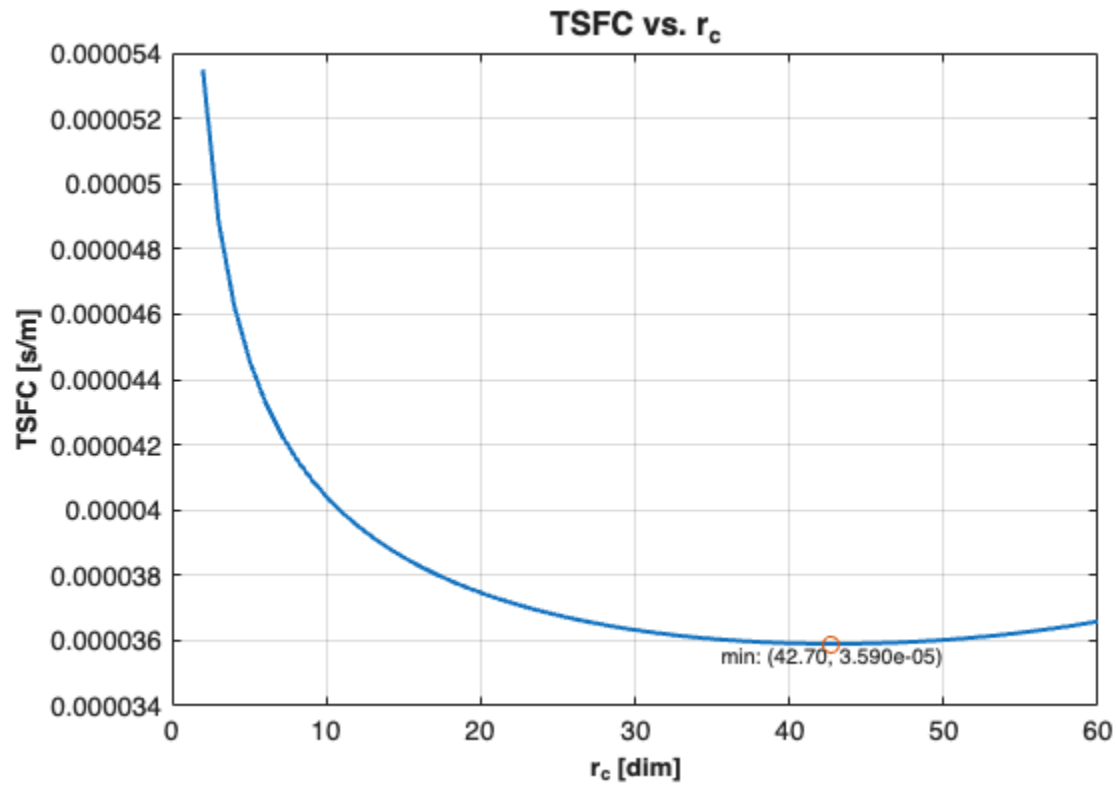


Figure 3: Efficiencies vs. r_c

```
figure('Position', [200, 200, 600, 400]);
plot(rc, eta_th, 'LineWidth', 1.5, 'DisplayName', '\eta_{th}');
hold on;
plot(rc, eta_0, 'LineWidth', 1.5, 'DisplayName', '\eta_{o}');
plot(rc, eta_p, 'LineWidth', 1.5, 'Color', 'k', 'DisplayName', '\eta_p');
hold off;
legend('Location', 'southeast', 'FontSize', 9);
xlabel('r_c [dim]', 'FontWeight', 'bold', 'FontSize', 10);
ylabel('Efficiency', 'FontWeight', 'bold', 'FontSize', 10);
title('Efficiencies vs. r_c', 'FontSize', 12);
grid on;
```

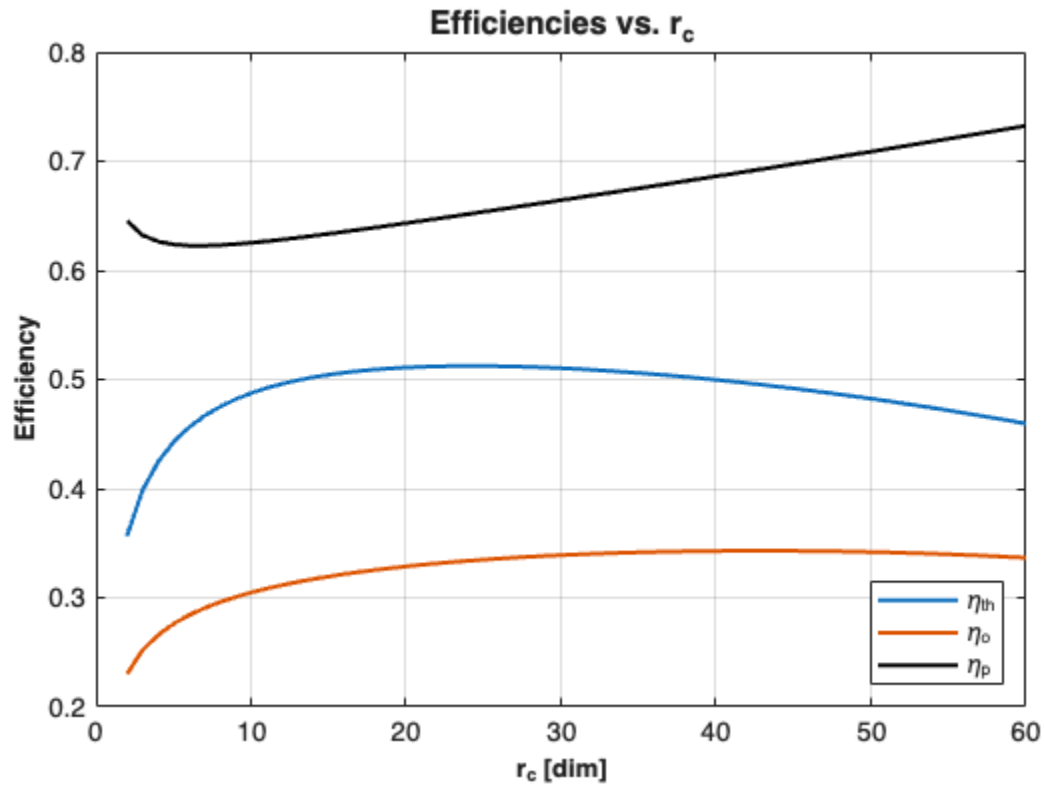
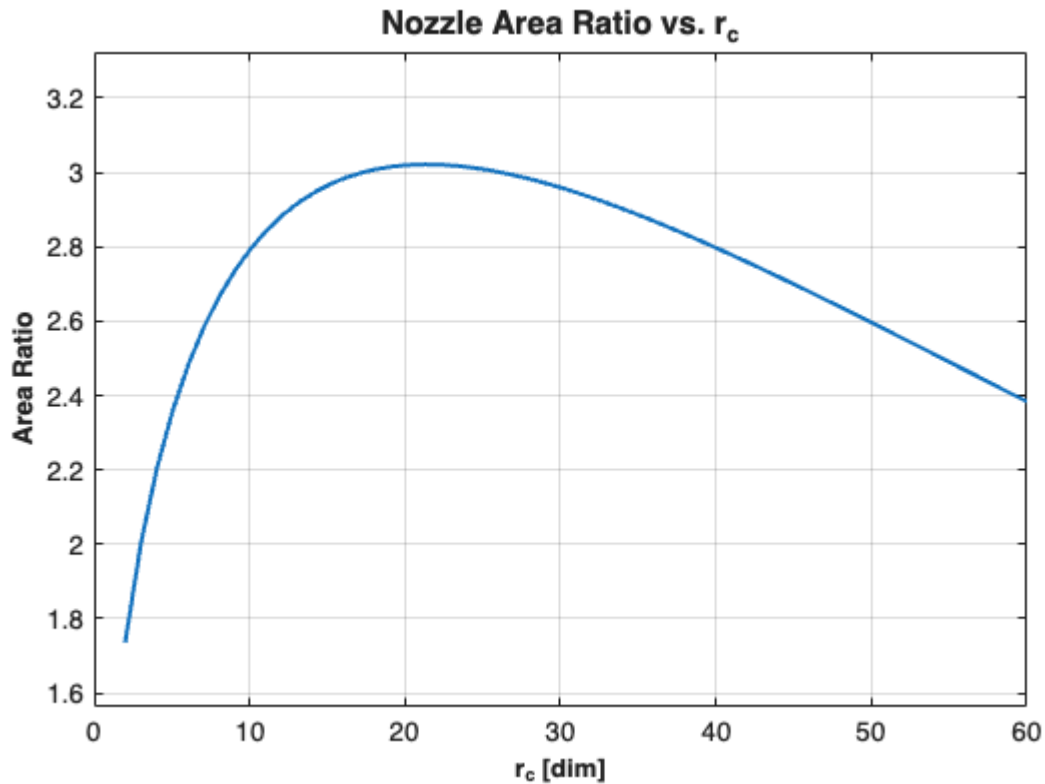


Figure 4: Nozzle Area Ratio vs. r_c

```
figure('Position', [250, 250, 600, 400]);
plot(rc, Ratio, 'LineWidth', 1.5);
ylim([0.9 * min(Ratio), 1.1 * max(Ratio)]);
xlabel('r_c [dim]', 'FontWeight', 'bold', 'FontSize', 10);
ylabel('Area Ratio', 'FontWeight', 'bold', 'FontSize', 10);
title('Nozzle Area Ratio vs. r_c', 'FontSize', 12);
grid on;
```



Save Data to CSV

```
data = table(rc, f_b, T03, T04, T05, P04, P05, T7, Me, Ue, W, I, TSFC,  
eta_th, eta_p, eta_0, Ratio, ...  
    'VariableNames', {'rc', 'f_b', 'T03', 'T04', 'T05', 'P04', 'P05', 'T7',  
    'Me', 'Ue', 'W', ...  
    'I', 'TSFC', 'eta_th', 'eta_p', 'eta_0', 'AreaRatio'});  
writetable(data, 'HW7_Q1_Data.csv');  
disp('Results saved to HW7_Q1_Data.csv');
```

Results saved to HW7_Q1_Data.csv

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